INSTITUTE OF MATHEMATICAL SCIENCES, FACULTY OF SCIENCE, UNIVERSITI MALAYA
SEMESTER 2, SESSION 2025/2026
TUTORIAL 7

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1. Determine whether the following functions are linearly independent or not by using the definition for the linearly independent functions.
   a. e^x, e^{x+4}
   b. e^x, e^{2x}
   c. $2x^2 + 8x,\ x^2 + 4x.$

2. For the following functions and differential equations:
   a. Show that the functions are solutions to the given differential equations.
   b. Verify that the functions are linearly independent using Wronskian determinant.
      i. $e^x, e^{-2x},\ y'' + y' - 2y = 0$
      ii. $x^2, x^{-1},\quad x^2y'' - 2y = 0$

3. Find the general solution of the given differential equation.
   a. $4y''+y' = 0$
   b. $y'' + 8y' + 16y = 0$
   c. $y'' + 9y = 0$
   d. $3y'' + 2y' + y = 0$
   e. $\frac{d^3u}{dt^3} + \frac{d^2u}{dt^2} - 2u = 0$

4. Solve the following initial value problems
   a. $y'' + 16y = 0,\qquad y(0) = 2,\ y'(0) = -2$
   b. $\frac{d^2y}{dt^2} - 4\frac{dy}{dt} - 5y = 0,\qquad y(1) = 0,\ y'(1) = 2$
   c. $y''' + 12y'' + 36y' = 0,\qquad y(0) = 0,\ y'(0) = 1,\ y''(0) = -7$